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- Introspection - William James (1800s) [Subjective]

- Psychology
  ↳ Study of "Psyche"
  ↳ Mind

Experimental Psychology
Social Psychology - How groups behave
Physiological Psychology - What brain areas are important for some behaviour
Cognitive Psychology - How knowledge influences behaviour
↳ To think and know knowledge

Disadvantages of Introspection :
Inconsistent
Individualistic - Vary from person to person

∴ Next Method : Behaviourism (objective) - 1900s - Watson
Basis for cognition : Behaviour

Stimulus (Input) $\xrightarrow{\text{Association}}$ Response (Output)

(i) Avoidance Behaviour
(ii) Approach Behaviour
} Response

Stimuli can be any input

Cognition
(Genes) wired
learnt (Multiple Stimuli ↔ Response)

Cognition
How do we know what we know

Repeated Association → Habit
There is nothing called mind

Pass the Salt

Sass the Palt

Salt, Please ! (Meaning mediating)

Could you pass the salt ?

Next Method : Cognitive Revolution (1950 - 60's)

- Cognitivism
- Tolman , Miller

  → Capacity limitations [7 ± 2 chunks]

  (Working memory)

  Temporary buffer

- Chomsky

  Associations doesn't explain how children learn language

  Universal Grammar (UG)

- Counterfactual reasoning

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→ Cocktail Party Problem ⇒ Forceful Attention

→ Conciousness - (i) Alive , Awake and Ability to respond to Stimuli

(ii) Ablity to experience and reflect on experiences (Polia)

Self - Concious ⇒ Humans

4^{rth} Epoch : Concious

→ Brain - basis of behaviour :

Neural Basis for Cognition

Spirit Basis ?

Avg. Response time : 100 ms

Temporal Lobe : Language

How do we study cognition ?

Behaviour
- Cognitive Psychology
- Experiments

  (BRED course)

  → (BRSM course)

Brain
- Cognitive Neuroscience
  - Healthy

Non-Invasive	Invasive
FMRI, MEG, EEG	PET, ECoG

  - Unhealthy

    ① Accidents

    Brain Injury, Disease

- Broca's Aphasia
  (Expressive Speech Disorder) - Frontal Lobe
- Wernicke's Aphasia
  (Comprehensive Speech Disorder) - Temporal Lobe

- Capgras Syndrome
  - Recognition
  - Memory
  - Comparison
  - Emotion Evaluation

  Integration

Amygdata : Fearful Stimuli

I - Behaviour
II - Brain
III - Computational Model
  Mathematical Model ⟷ Experiment
  Complex functionalities

Brain : Product of Evaluation

[INCM]

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→ <u>Neural</u> <u>Basis</u> :

<u>Meninges</u>
Pia Matter

Top (Superior/Rostral)

frart
(Anterior)

Back (Posterior)

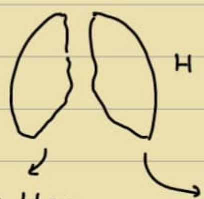
Cerebellum

Cerebrum

Bottom (Caudal)

Spinal Cord

L



H

Two Hemispheres :
LH and RH

Gray Matter

White Matter

Coronal Section

Vertebral bones protect Spinal Cord.
Fluid sorrounds this is Cerebrospinal Fluid (CSF)

- Skull Bone

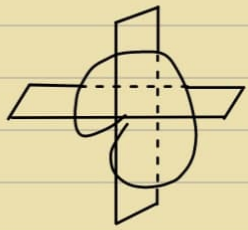
Gray Matter : Neural Cell body
White Matter : Fibres / Axons
(White coating) that connect

- Central Nervous System :
  Cerebellum + Spinal Cord
  Recieves processes and gives out output

- Peripheral Nervous System :
  Generates signal for muscles to contract

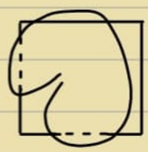
- Dopamine : Pleasure

- <u>Sections of the brain</u> :



Coronal → RH and LH

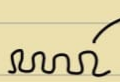
Horizontal (Transverse) ↗



(Mid - Sagittal)

Ventral : Front
Dorsal : Back

Cortex :



gyrus (bump)

Sulkus (valley)
[Groove]

Big Grooves are called Fissures

Central Sulkus : Seperates front and back
Lateral Fissure : Seperates top and bottom
Horizontal Fissure : Seperates left and right (Longitudinal)

Transverse Fissure : Seperates Cerebrum and Cerebellum

- 4 Lobes :

(1) Frontal Lobe : Cerebrum : Neocortex

- Well developed
- Newest part of Neocortex (42)
- Keeps developing until 21 years
- Working Memory , Executive function (Out-of-Sight, Out-of-mind)
- Planning
- Emotional Evaluation (Phineas Gage), Personality
- Decision Making
- Motor Cortex
  - Speech (Brocas Area)

(2) Paretal Lobe :

- Somatosensory cortex
- Proprioception : Sense of where their body parts are
- Goal-directed Action
  (Spatial Orientation)

- CNS and PNS work together

(3) Occipital Lobe :

- Visual Cortex : Vision
  Object Recognition, Object Identification
  Color, Shape, Size
  Object Motion (Movement of Object)

(4) Temporal Lobe :

- Object Naming
- Language
- Speech Comprehension (Verniki's Area)
- Auditory Processing (Auditory Cortex)
  Input from ears

- Subcortical Struct
  - Amygdala (Fear)
  - Hippocampus (Long-Term Memory)
  - Basal Gangha (Motor)
  - Thalamus (Gathers incoming things and distribute to appropriate)
    (i.e. Sensory Exchange - Waystation for Sensory

- Cerebellum
  - Motor Coordination
  - Compromised for Long-Term Alcoholics

- Hypothalamus (Mid-Brain / Old-brain)
  - Hunger
  - Breathing control / Respiration control
  - Temp. Control
  - Sleep-Wake Cycle

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- Dementia
- Functional MRI
- Coma : Vegetative State

- Reptilian Complex

$\Downarrow$

Limbic system : Happy, sad, etc

$\Downarrow$

Neocortex

- EEG : Electrode Cap

  Deep Sleep : Low Frequency, High Altitude

  Fully Awake : High Frequency, Low Altitude

- fMRI : Response $\downarrow$ when visualising

  Blood Oxygenation / Deoxygenation

  Brain Computer Interface

  Avg. Connectivity in Neural System : 10^4

→ Deep Brain Simulation - Parkinson's disease

Action Potential - 10^{-3} s

Nature & Nurture

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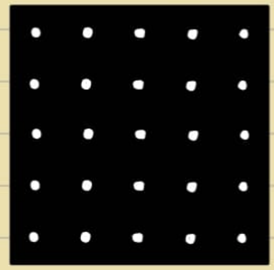
→ Perception : Interpretation of raw input data

- Neurons - Electrically excitable
  Axon - Carry information

Visual Illusion :



- Synapses - Electrochemical Majorly
  Gap Junction - Reduced Evolutionarily

- Hodgkin and Huxley - Squid
  Resting : −70 V
  Threshold : − 55 V

- Inter - Pulse Intervals

- Action Potentials to Perception :
  Modularity

- Grandmother Coding - Localist theory
  i.e. one neuron corresponds to face of Grandmother

Multimodal distribution - Distributed Processing

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Input → Memory → Output

Thought Process : Reasoning and Decision Making

- Human : Sequential (Speech) &
  Parallel (Speech + Vision)

Attention - Sustained Attention
- Divided Attention

Desire gives ability to Plan, Execute and Fulfills

- Emotion : Sense of Morality, Empathy

Perception : Interpretation

- Top down : Allows you to determine what is unclear / Ambiguous
  Bottom Up : See the illusion directly

- Short Term Memory - Hours (1 hr, 2 hrs)
  Procedural Memory : Motor Memory
  Priming ≡ Memory
  ↪ Prepotentiation of our system so that in future, we can see patterns quickly

- Homunculus : Declarative Memory
  Coronal Section

Fine grip - Not in animals, only in Humans